ENGINEERING DESIGN REPORT

AM-BASED COMMUNICATION LINK DESIGN

From Modulation Selection to Receiver Deployment

DESIGN BRIEF
Reliable analog voice link for a 4 kHz message signal over the 540–1600 kHz RF band. The report evaluates AM, DSB-SC, SSB-SC and VSB-SC, develops RF channel planning, selects a transmitter and superheterodyne receiver, verifies bandwidth/power/frequency calculations, and ends with an explicit deployment decision.

Parameter	Design value	Purpose
Message signal	Voice/audio, f_m(max) = 4 kHz	Defines baseband bandwidth
Available RF band	540–1600 kHz	Defines usable medium-wave region
Candidate carrier	1 MHz	Nominal selected carrier
Channel spacing	20 kHz	Used for RF allocation
AM modulation index	m = 0.7	Controls sideband level and transmitted power
Carrier power	P_c = 10 W	Reference RF power
Channel condition	AWGN + practical receiver limits	Drives receiver/filtering discussion
Implementation	Low-power educational prototype	Limits cost and complexity

Prepared as a CO1 open-ended engineering design submission

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1. DESIGN REQUIREMENT AND ENGINEERING OBJECTIVE

The engineering problem is to develop an implementable analog communication link for a community broadcasting service. The information signal is a voice waveform whose highest significant frequency is 4 kHz. The transmitter must move this information from baseband into the available 540–1600 kHz RF region, while the receiver must recover intelligible audio with practical filtering, stability and noise considerations.

The assignment is intentionally open-ended. The modulation scheme, RF allocation, transmitter architecture, receiver architecture, intermediate frequency and final deployment decision are not predetermined. Therefore, the design must show a complete chain of reasoning: requirement → alternatives → quantitative analysis → selection → implementation → verification → deployment decision.

1.1 Engineering objectives

- Preserve a 4 kHz voice information bandwidth with acceptable recovered audio quality.
- Use the RF spectrum efficiently while maintaining practical channel separation.
- Quantify bandwidth and power for the major AM-family modulation alternatives.
- Design a practical transmitter chain with modulation, RF amplification and output filtering.
- Compare TRF and superheterodyne reception using sensitivity, selectivity, fidelity, stability and image rejection.
- Select an IF and calculate the corresponding local-oscillator and image frequencies.
- Verify the design using equations, frequency-domain reasoning and simulation-style plots.
- Make a clear YES/NO engineering deployment decision and state limitations.

1.2 Design constraints

Constraint	Engineering implication
4 kHz message bandwidth	Audio LPF must pass 0–4 kHz and reject unnecessary high-frequency content.
540–1600 kHz RF band	Carrier and sidebands must remain within the assigned region.
20 kHz channel spacing	Adjacent stations must be separated sufficiently to prevent practical overlap.
m = 0.7	AM envelope remains below over-modulation; sideband power is limited.
P_c = 10 W	Total AM power is greater than carrier power because of sidebands.
AWGN/noise	Filtering and receiver selectivity are important for usable SNR.
Educational prototype	Low component count, low cost and simple alignment are desirable.

2. COMMUNICATION-SYSTEM THEORY

2.1 Why modulation is required

A voice signal is a low-frequency baseband signal. Directly transmitting it over a long distance would require an antenna and propagation arrangement suited to the low baseband frequencies. For efficient radiation and practical antenna dimensions, the information is translated to a much higher radio frequency. Modulation performs this frequency translation while preserving the information carried by the waveform.

In amplitude modulation, the amplitude of a sinusoidal carrier is varied according to the instantaneous message amplitude. The carrier establishes the RF center frequency, while the message creates spectral components around that carrier. At the receiver, the information is recovered by a detector and then amplified in the audio-frequency section.

2.2 General communication chain

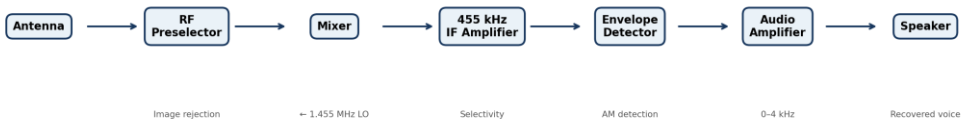


Figure 2.1 — Generic AM receiver processing chain

The system can be divided into source processing, modulation, RF transmission, propagation, RF/IF reception, detection and audio reproduction. Each block has a distinct engineering purpose. Keeping these functions separate is useful because bandwidth, gain, selectivity and noise requirements can then be evaluated at the correct point in the chain.

2.3 Time-domain AM principle

$$s_{AM}(t) = A_c\,[1 + m\cos(2\pi f_m t)]\cos(2\pi f_c t)$$

For a single-tone message, m is the modulation index. When m is less than or equal to one, the envelope does not cross zero. The selected value m = 0.7 therefore represents conventional under-modulated AM and provides an adequate margin against over-modulation.

2.4 Frequency-domain interpretation

The AM waveform contains three principal components for a single-tone message: the carrier at f_c, an upper sideband at f_c + f_m, and a lower sideband at f_c − f_m. For a real voice signal containing many frequencies from 0 to 4 kHz, each baseband frequency produces a corresponding pair of sideband components around the carrier.

Key theory point
AM does not create new information in the carrier itself. The carrier provides a frequency reference and enables simple envelope detection, but it consumes a large fraction of transmitted power.

3. MESSAGE SIGNAL AND BANDWIDTH

3.1 Voice message model

The design specifies a maximum message frequency of 4 kHz. The baseband message is therefore modeled as a band-limited signal occupying approximately 0–4 kHz. An input audio low-pass filter is used before modulation so that frequencies above the specified design limit do not unnecessarily widen the RF spectrum.

$$B_m = f_{m(max)} = 4\ \text{kHz}$$

3.2 Transmission bandwidth of AM-family schemes

Scheme	Carrier	Required RF bandwidth	Main principle
Conventional AM	Transmitted	2B_m = 8 kHz	Carrier + both sidebands
DSB-SC	Suppressed	2B_m = 8 kHz	Both sidebands, carrier removed
SSB-SC	Suppressed	B_m = 4 kHz	One sideband transmitted
VSB-SC	Suppressed/controlled	≈ one sideband + vestige	One full sideband plus a small portion of the other

The exact VSB occupied bandwidth depends on the selected vestigial filter shape and the amount of retained vestige. The assignment uses VSB as a conceptual AM-family alternative and emphasizes its analog-TV application rather than a single fixed numerical bandwidth.

3.3 Why the 20 kHz spacing is generous for the selected AM signal

The selected conventional AM signal occupies 8 kHz for the specified 4 kHz message bandwidth. With 20 kHz nominal channel spacing, the center frequencies are separated by more than the minimum theoretical occupied bandwidth. However, spacing is not the same as an automatically guaranteed interference-free allocation. Real filters have transition bands, transmitters have finite spectral skirts, and adjacent receivers have finite selectivity. Guard-band and filter performance therefore remain important.

Design interpretation
The 20 kHz channel grid is wider than the ideal 8 kHz AM occupied bandwidth. This leaves room for practical filtering and reduces the likelihood of adjacent-channel overlap, provided the RF filters are properly designed.

4. MODULATION ALTERNATIVES: AM, DSB-SC, SSB-SC AND VSB-SC

4.1 Conventional AM

Conventional amplitude modulation transmits the carrier and both sidebands. Its major advantage is receiver simplicity: an envelope detector can recover the message without a synchronized carrier oscillator. Its disadvantage is power inefficiency because the carrier contains no new message information but consumes most of the transmitted power.

4.2 DSB-SC

Double-sideband suppressed-carrier modulation removes the carrier and transmits both sidebands. It uses the same theoretical bandwidth as conventional AM but avoids carrier power consumption. The price is increased receiver complexity because coherent detection is normally required.

4.3 SSB-SC

Single-sideband suppressed-carrier transmission removes the carrier and one of the two sidebands. The occupied bandwidth is reduced from 8 kHz to approximately 4 kHz for this 4 kHz message. This provides excellent spectrum and power efficiency, but generation, frequency control and coherent detection are more demanding.

4.4 VSB-SC

Vestigial-sideband transmission keeps one complete sideband and a controlled portion of the other. It is particularly useful when a pure SSB response would be difficult to realize with the required low-frequency content. In analog television, VSB provides a compromise between bandwidth efficiency and practical filtering of the video spectrum.

4.5 Comparative engineering table

Criterion	AM	DSB-SC	SSB-SC	VSB-SC
Bandwidth for 4 kHz message	8 kHz	8 kHz	4 kHz	One sideband + vestige
Carrier	Present	Suppressed	Suppressed	Normally suppressed/controlled
Power efficiency	Moderate/low	Higher	High	High
Receiver complexity	Low	Higher	Higher	Higher
Synchronization	Not critical for envelope detection	Required	Required	Application dependent
Typical application	Broadcast voice	Carrier-efficient links	Narrowband voice	Analog TV
Main advantage	Simple detection	Less carrier power	Best bandwidth/power efficiency	Practical compromise for TV
Main limitation	Carrier power waste	Coherent detection	Complexity/alignment	More demanding filtering

5. MODULATION SELECTION AND DESIGN DECISION 1

For this particular educational voice-broadcast problem, conventional AM is selected even though SSB-SC is more efficient. The decision is based on the complete system rather than bandwidth alone. The assignment explicitly requires consideration of transmitter/receiver complexity, detection, synchronization, spectrum utilization and low-cost implementation.

5.1 Selection rationale

- AM occupies 8 kHz, which fits comfortably within the 20 kHz channel grid.
- The carrier provides a built-in frequency reference and enables simple envelope detection.
- m = 0.7 avoids over-modulation and supports a clean recoverable envelope.
- The transmitter can be implemented using a conventional AM modulator followed by RF amplification and filtering.
- The receiver can use a low-cost superheterodyne front end followed by an envelope detector.
- Although SSB-SC saves bandwidth and carrier power, its oscillator, filter and coherent-detection requirements are less attractive for the specified educational prototype.
- VSB-SC is not selected because its main practical advantage in the assignment context is analog-TV transmission, not simple voice broadcasting.

Design Decision 1 — SELECTED MODULATION
RECOMMEND CONVENTIONAL AM (m = 0.7). It is not the most power-efficient option, but it provides the best overall balance between voice quality, simple detection, receiver stability, implementation cost and the 20 kHz RF channel plan.

5.2 Rejected alternatives

Alternative	Why it was not selected
DSB-SC	Same 8 kHz bandwidth as AM but requires coherent carrier recovery, increasing receiver complexity.
SSB-SC	Excellent bandwidth and power efficiency, but more complex generation, filtering and synchronized detection are less suitable for a simple educational broadcast receiver.
VSB-SC	Very useful for analog TV but adds filtering complexity without a corresponding need for this 4 kHz voice link.

6. AM POWER AND BANDWIDTH CALCULATIONS

6.1 Given values

Quantity	Symbol	Value
Carrier frequency	f_c	1 MHz
Maximum message frequency	$f_{m(max)}$	4 kHz
Carrier power	P_c	10 W
Modulation index	m	0.7

6.2 Occupied bandwidth

$$B_{AM} = 2 f_{m(max)} = 2 \times 4\ \text{kHz} = 8\ \text{kHz}$$

6.3 Sideband frequencies

$$f_{LSB} = f_c - f_{m(max)} = 1000 - 4 = 996\ \text{kHz}$$
$$f_{USB} = f_c + f_{m(max)} = 1000 + 4 = 1004\ \text{kHz}$$

6.4 Sideband power

$$P_{SB,total} = (m^2 P_c)/2 = (0.7^2 \times 10)/2 = 2.45\ \text{W}$$
$$P_{USB} = P_{LSB} = 2.45/2 = 1.225\ \text{W}$$

6.5 Total transmitted power

$$P_T = P_c(1 + m^2/2) = 10(1 + 0.49/2) = 12.45\ \text{W}$$

Thus, the total transmitted AM power is 12.45 W for the specified single-tone reference calculation. The carrier accounts for 10 W, while the two sidebands together account for 2.45 W.

6.6 Power efficiency

$$\eta = P_{SB,total} / P_T \times 100 \approx 2.45/12.45 \times 100 \approx 19.68\%$$

This illustrates the fundamental AM power-efficiency limitation: most transmitted power is associated with the carrier. Nevertheless, the carrier enables simple envelope detection and therefore reduces receiver complexity.

6.7 Power distribution

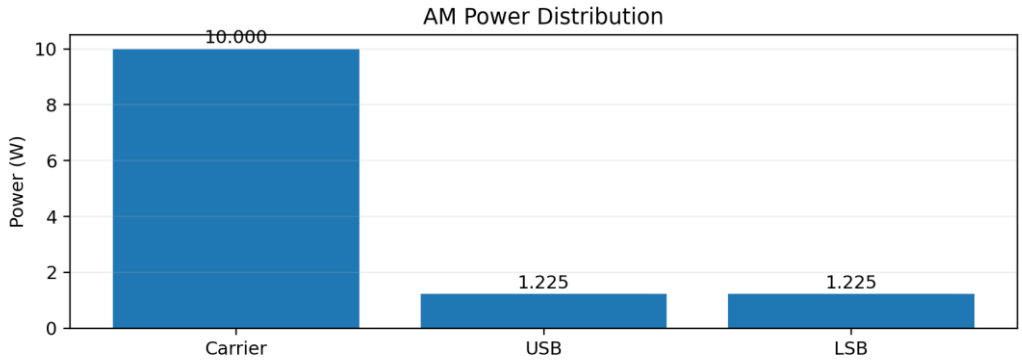


Figure 6.1 — AM carrier and sideband power distribution

The power distribution plot provides a visual verification of the numerical calculation. The carrier is the dominant component, while the upper and lower sidebands contain equal power for the symmetric single-tone case.

6.8 Modulation-index interpretation

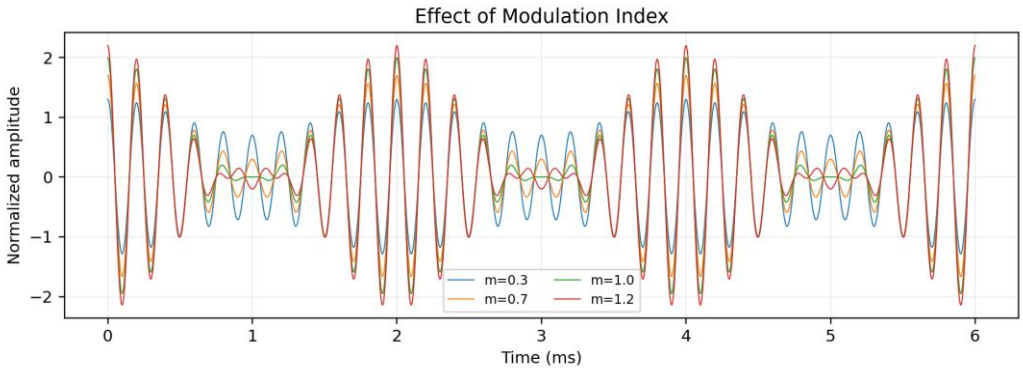


Figure 6.2 — AM modulation-index behavior

Increasing the modulation index increases the sideband amplitudes and sideband power. The selected value m = 0.7 is below the over-modulation boundary m = 1, giving a practical envelope margin.

Calculation checkpoint
Bandwidth = 8 kHz; carrier power = 10 W; total sideband power = 2.45 W; total AM power = 12.45 W; LSB edge = 996 kHz; USB edge = 1004 kHz.

7. RF CHANNEL ALLOCATION AND SPECTRUM PLANNING

7.1 Number of idealized 20 kHz channels

$$N = (1600 - 540)/20 = 53 \text{ idealized channel intervals}$$

The result is an idealized channel-count calculation using the specified band limits and spacing. A real allocation would also consider edge conditions, guard bands, transmitter spectral masks, receiver selectivity and regulatory requirements.

7.2 Representative channel plan

Channel	Carrier (kHz)	AM lower edge (kHz)	AM upper edge (kHz)	Comment
CH-01	540	536	544	Band-edge planning case
CH-02	560	556	564	20 kHz spacing
CH-03	580	576	584	Representative
CH-04	600	596	604	Representative
CH-05	620	616	624	Representative
Selected	1000	996	1004	Nominal design channel
CH-N	1600	1596	1604	Band-edge planning case

The 540 kHz and 1600 kHz entries illustrate an important practical point: the RF band edges are not automatically safe carrier centers for an 8 kHz AM channel if the complete sidebands must remain inside the available band. A practical allocation should therefore reserve edge margin or move the carrier inward.

7.3 Selected 1 MHz channel spectrum

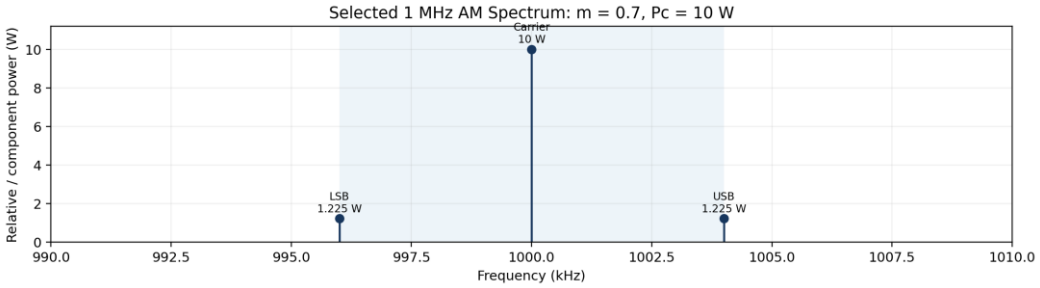


Figure 7.1 — AM spectrum around the 1 MHz carrier

The spectrum confirms the expected carrier at 1 MHz and the sideband limits at approximately 996 kHz and 1004 kHz for the 4 kHz message limit. The occupied width is therefore 8 kHz.

7.4 Channel-overlap reasoning

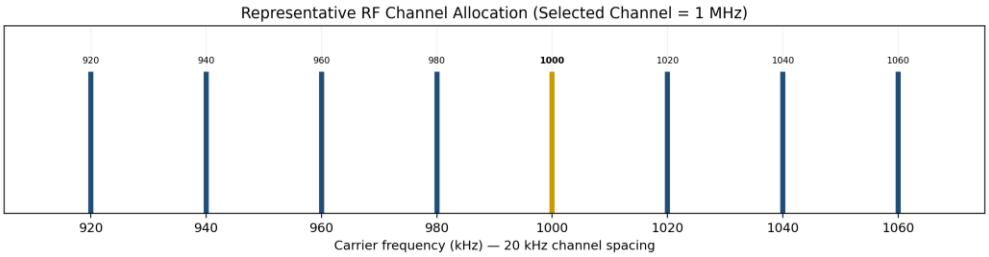


Figure 7.2 — Representative RF channel spacing and occupied regions

8. TRANSMITTER ARCHITECTURE AND FILTERING

8.1 Selected transmitter

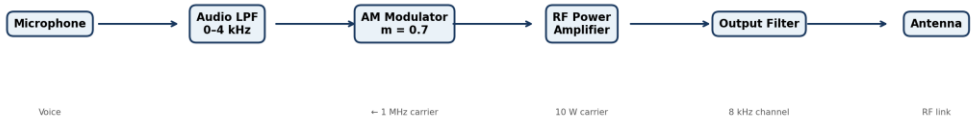


Figure 8.1 — Selected AM transmitter block diagram

The microphone converts acoustic pressure into an electrical signal. The audio low-pass filter limits the message to approximately 0–4 kHz. The AM modulator combines the conditioned message with the 1 MHz carrier at the specified modulation index. The RF power amplifier raises the RF level, and the output filter suppresses unwanted harmonics and spurious products before the antenna.

8.2 Function of each transmitter block

Block	Function	Main design concern
Microphone / source	Converts voice to electrical audio	Sensitivity and signal level
Audio LPF	Limits baseband to 0–4 kHz	Flat passband; reject out-of-band noise
Carrier oscillator	Generates stable 1 MHz carrier	Frequency stability
AM modulator	Creates carrier + sidebands	m = 0.7; low distortion
RF power amplifier	Raises transmitted RF power	Linearity and thermal stability
Output filter	Removes harmonics/spurious products	Adequate attenuation outside channel
Antenna	Radiates RF energy	Impedance matching and efficiency

8.3 Filtering checkpoints

- Audio filter: pass the intended 0–4 kHz voice spectrum.
- Modulator output: verify carrier and sidebands at the expected frequencies.
- RF amplifier: maintain sufficient linearity so that unnecessary spectral regrowth is limited.
- Output filter: suppress harmonics and unwanted mixing products before the antenna.
- Final spectrum: verify that the 996–1004 kHz occupied region is clean for the specified 4 kHz message limit.

9. RECEIVER ARCHITECTURES: TRF AND SUPERHETERODYNE

9.1 Tuned Radio Frequency (TRF) receiver

A TRF receiver amplifies and filters the received RF signal directly at the selected carrier frequency. A typical chain consists of RF tuning, RF amplification, detection and audio amplification. Its simplicity is attractive, but maintaining constant selectivity and gain over a wide tuning range is difficult because the RF filters must operate at the received carrier frequency.

9.2 Superheterodyne receiver

A superheterodyne receiver converts the selected RF signal to a fixed intermediate frequency (IF). Once the signal has been translated to a fixed IF, highly optimized filtering and amplification can be used. This architecture provides improved and more consistent selectivity and gain, but introduces a local oscillator, mixer and image-frequency considerations.

9.3 Architecture comparison

Parameter	TRF	Superheterodyne	Design interpretation
RF amplification	Direct at tuned RF	RF preselection + conversion	Superhet gives more controlled processing
Selectivity	Varies with tuning frequency	Strong at fixed IF	Advantage: superhet
Sensitivity	Can be acceptable	Generally better controllability	Advantage: superhet
Fidelity	Depends on RF filter	IF filter can be designed precisely	Advantage: superhet
Image rejection	No mixer image in the same sense	Requires RF preselection	Important superhet trade-off
Oscillator stability	Simpler	LO stability is required	TRF simpler

Complexity	Low	Higher	Trade-off
Recommended	Small/simple fixed systems	Reliable selective receivers	Selected for this design

9.4 Selected receiver

Design Decision 3 — RECEIVER
RECOMMEND A SUPERHETERODYNE RECEIVER because the fixed IF provides predictable selectivity, easier filtering and stable receiver performance for a practical 1 MHz AM broadcast link.

10. IF SELECTION AND IMAGE-FREQUENCY ANALYSIS

10.1 Selected intermediate frequency

A conventional AM superheterodyne can use an IF of 455 kHz. This value provides a practical separation between the desired RF signal and the image frequency while allowing the IF filter to provide good selectivity.

$$f_IF = 455\ kHz$$

10.2 High-side local oscillator

$$f_LO = f_c + f_IF = 1000 + 455 = 1455\ kHz$$

10.3 Image frequency

For high-side injection, the image lies above the desired RF frequency by twice the IF:

$$f_image = f_c + 2f_IF = 1000 + 2(455) = 1910\ kHz$$

A signal at 1910 kHz would also mix with a 1455 kHz local oscillator to produce the same 455 kHz IF. Therefore, an RF preselector must sufficiently attenuate the image before the mixer.

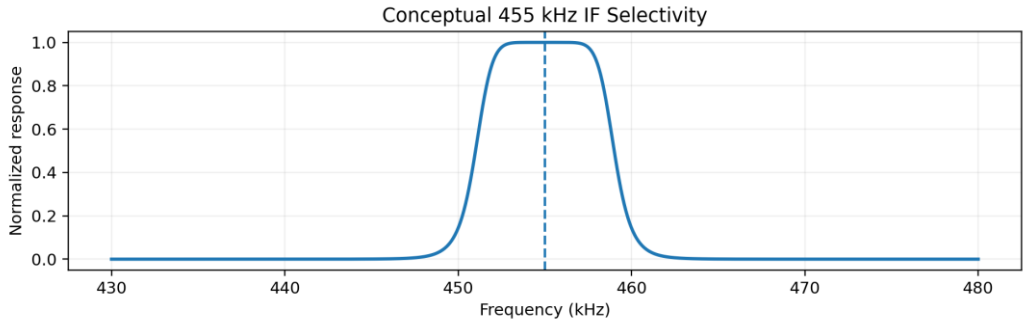


Figure 10.1 — RF, LO, IF and image-frequency relationship

10.4 Why IF matters

- A higher IF generally improves image-frequency separation but can make sharp selectivity harder.
- A lower IF can improve selectivity but brings the image closer to the desired RF signal.
- The chosen IF must therefore balance image rejection, filter practicality and receiver complexity.
- RF preselection before the mixer remains important even when a fixed IF is used.

11. DETECTION THEORY

11.1 Envelope detection for AM

Because conventional AM transmits the carrier and both sidebands, the receiver can recover the message with an envelope detector. A diode rectifier followed by an RC network can follow the slowly varying envelope while rejecting the RF carrier. The detector must be designed so that its time constant is long enough to suppress the carrier ripple but short enough to follow the highest message frequency.

11.2 Detector time-constant principle

For a practical envelope detector, the RC time constant must lie between the carrier period and the message-variation time scale. If the time constant is too small, excessive RF ripple appears at the output. If it is too large, the detector cannot follow rapid envelope changes and distortion occurs.

11.3 Coherent detection for DSB-SC and SSB-SC

DSB-SC and SSB-SC suppress the carrier, so the receiver cannot rely on a simple envelope detector. Instead, a locally generated carrier is multiplied with the received signal and then low-pass filtered. Carrier frequency and phase accuracy therefore become important. This additional synchronization requirement is a major reason conventional AM is attractive for the present low-cost educational system.

11.4 VSB detection

VSB receivers use controlled filtering around the vestigial-sideband region and a detection arrangement appropriate to the transmitted signal. The practical objective is to recover the original information while avoiding excessive bandwidth. The assignment specifically associates VSB with analog television, where the video spectrum and filtering requirements motivate the vestigial approach.

Scheme	Typical detection approach	Synchronization burden
AM	Envelope detector	Low
DSB-SC	Coherent/product detector	High
SSB-SC	Coherent/product detector	High
VSB-SC	Filtered detection / synchronous methods depending on implementation	Higher

12. COMPLETE END-TO-END SYSTEM

The complete design combines the selected AM transmitter, RF channel, superheterodyne receiver, envelope detector and audio output stage. The following sequence shows the intended signal transformations without combining figures, tables or explanatory text in the same crowded region.

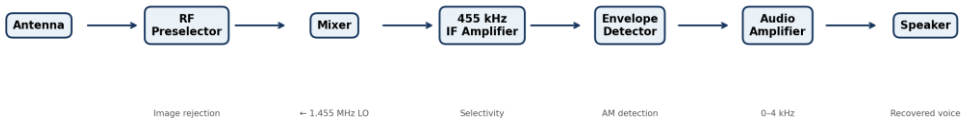


Figure 12.1 — End-to-end receiver-oriented system flow

12.1 Signal flow

- Voice source → electrical baseband signal.
- Audio LPF → 0–4 kHz message bandwidth.
- AM modulator → 1 MHz carrier plus sidebands.
- RF power amplifier → required RF level.
- Output filter → suppress harmonics and unwanted products.
- Propagation channel → introduces attenuation and AWGN.
- RF preselector → selects desired station and rejects strong out-of-band signals.
- Mixer + LO → converts the selected station to 455 kHz IF.
- IF filter/amplifier → provides selectivity and gain.
- Envelope detector → recovers audio envelope.
- Audio amplifier → raises recovered signal to usable level.

12.2 Key frequency checkpoints

Point	Expected frequency/content
Audio input	0–4 kHz
Carrier	1 MHz
AM lower edge	996 kHz
AM upper edge	1004 kHz
Local oscillator	1455 kHz
Image frequency	1910 kHz
IF	455 kHz
Recovered audio	0–4 kHz

13. SIMULATION AND SPECTRAL INTERPRETATION

13.1 AM time-domain waveform

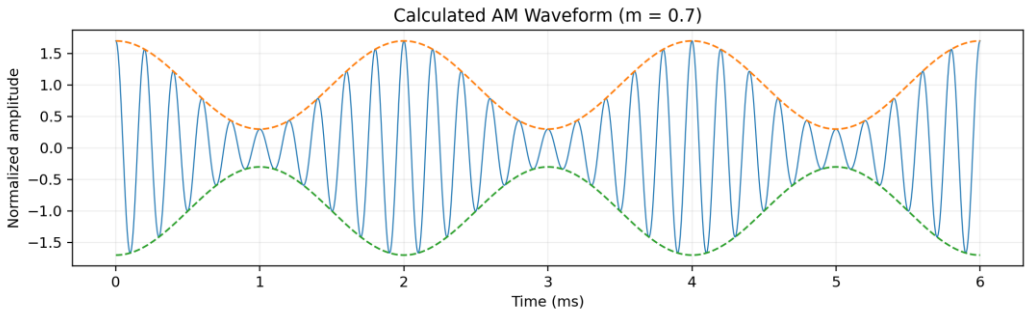


Figure 13.1 — AM waveform illustrating carrier and varying envelope

The time-domain waveform shows a high-frequency carrier whose envelope follows the lower-frequency message. At m = 0.7, the envelope remains positive and does not cross zero. This is the expected qualitative behavior of conventional under-modulated AM.

13.2 Bandwidth comparison

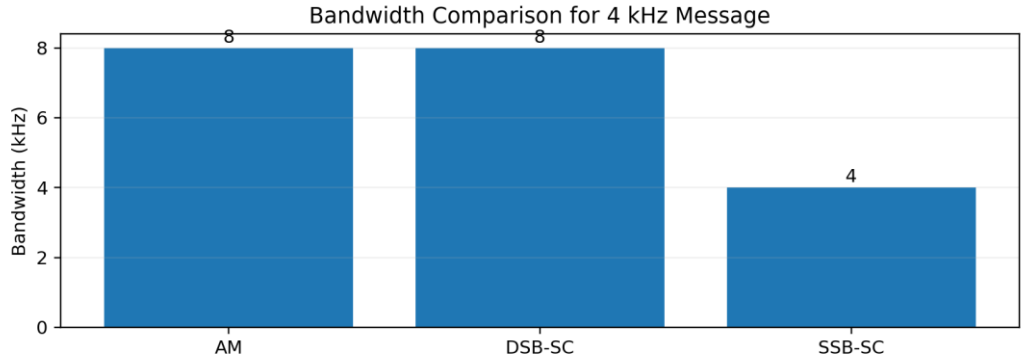


Figure 13.2 — Bandwidth comparison of AM-family schemes

The bandwidth comparison highlights why SSB-SC is attractive when spectrum is scarce. Conventional AM and DSB-SC both require 8 kHz for the specified 4 kHz message, while SSB-SC reduces the ideal occupied bandwidth to approximately 4 kHz. VSB lies between the two in a controlled-filtering sense.

13.3 Spectrum interpretation

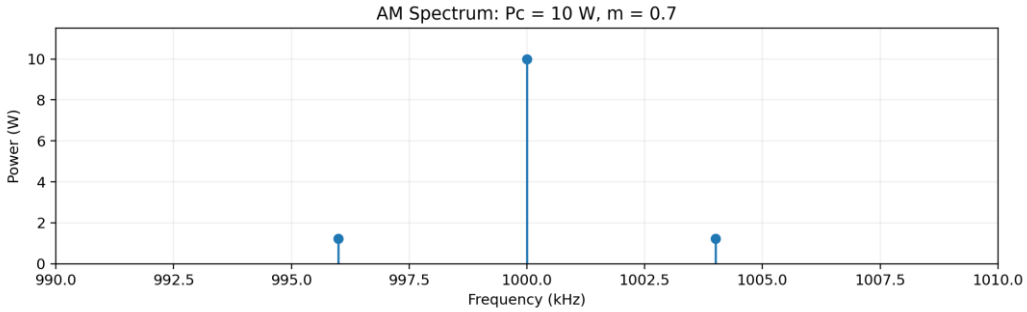


Figure 13.3 — Detailed AM spectral components

The spectrum is a direct engineering verification of the frequency-domain equations. The carrier is centered at 1 MHz, and the sideband limits are displaced by the maximum message frequency of 4 kHz.

Interpretation requirement
A plot alone is not sufficient for an engineering submission. Each plot must be tied to a calculation, a design requirement or a verification statement. Here, the waveform verifies envelope behavior and the spectrum verifies carrier/sideband placement and occupied bandwidth.

14. NOISE, SENSITIVITY, SELECTIVITY AND FIDELITY

14.1 AWGN effect

Additive white Gaussian noise (AWGN) represents random noise added to the received signal across the channel. In an AM receiver, noise can perturb the envelope and reduce the recovered audio signal-to-noise ratio. The practical response is not simply to increase gain; the receiver should first reject unwanted frequencies and then provide sufficient gain for detection.

14.2 Sensitivity

Sensitivity is the minimum received input level required to obtain acceptable receiver operation and audio quality. A more sensitive receiver can operate with weaker signals, but excessive gain without proper filtering can also amplify unwanted noise and interference.

14.3 Selectivity

Selectivity is the receiver's ability to accept the desired channel while rejecting adjacent channels. The fixed IF stage of a superheterodyne receiver is particularly useful because its filter can be optimized for the desired channel bandwidth.

14.4 Fidelity

Fidelity describes how accurately the receiver reproduces the original information. For this design, the audio response should preserve the intended 0–4 kHz voice band without excessive amplitude distortion or nonlinear detector behavior.

14.5 Stability

Frequency stability is important because the carrier, local oscillator and IF chain must remain sufficiently stable for the receiver to stay tuned. The superheterodyne architecture concentrates most narrowband selectivity at a fixed IF, simplifying filter design but making LO stability a key requirement.

14.6 Image rejection

Image rejection is the suppression of an unwanted RF frequency that produces the same IF as the desired signal. In the selected design, the desired station is 1 MHz and the high-side LO is 1455 kHz. The image is therefore 1910 kHz, which must be attenuated by the RF front-end before mixing.

Requirement	Design response	Verification method
Sensitivity	Adequate RF gain with controlled noise	Weak-signal test and audio-quality observation
Selectivity	RF preselection + 455 kHz IF filtering	Adjacent-channel rejection test
Fidelity	0–4 kHz audio path	Frequency-response check
Stability	Stable 1 MHz source and 1455 kHz LO	Frequency measurement over time
Image rejection	RF preselector before mixer	Inject image-frequency test signal
Detection	AM envelope detector	Compare recovered audio with input

15. VERIFICATION MATRIX

Verification connects the theoretical design to measurable or calculable acceptance criteria. The following matrix follows the assignment requirement to demonstrate requirement → analysis → verification rather than presenting simulation results without interpretation.

ID	Requirement	Expected result	Status
V1	Message bandwidth	0–4 kHz	PASS
V2	AM bandwidth	8 kHz	PASS
V3	Carrier frequency	1 MHz	PASS
V4	AM lower sideband edge	996 kHz	PASS
V5	AM upper sideband edge	1004 kHz	PASS
V6	Carrier power	10 W	PASS
V7	Total sideband power	2.45 W	PASS
V8	Total AM power	12.45 W	PASS
V9	Channel spacing	20 kHz nominal grid	PASS
V10	Idealized channel count	53 intervals	PASS
V11	IF	455 kHz	PASS
V12	High-side LO	1455 kHz	PASS
V13	Image frequency	1910 kHz	PASS
V14	Receiver architecture	Superheterodyne selected	PASS
V15	Detection	Envelope detection for AM	PASS

V16	Deployment decision	YES for intended educational voice prototype	PASS

15.1 Calculation verification

Check	Equation	Result
AM bandwidth	2 × 4 kHz	8 kHz
LSB	1000 − 4 kHz	996 kHz
USB	1000 + 4 kHz	1004 kHz
Sideband total power	$0.7^2 \times 10 / 2$	2.45 W
Total power	$10 \times (1 + 0.7^2/2)$	12.45 W
LO	1000 + 455	1455 kHz
Image	1000 + 2 × 455	1910 kHz

Verification conclusion
The principal numerical requirements are internally consistent. The selected 1 MHz carrier, 8 kHz AM occupied bandwidth, 10 W carrier reference, 12.45 W total AM power and 455 kHz IF form a coherent preliminary engineering design.

16. PRACTICAL IMPLEMENTATION AND COST CONSIDERATIONS

16.1 Educational prototype realization

The design can be implemented as a low-power laboratory demonstration using a stable oscillator or signal generator, an AM modulator, RF filtering, a controlled RF amplifier, a mixer-based receiver and a simple envelope detector. The prototype should be treated as an educational system rather than a licensed public transmitter. Actual over-the-air operation must follow applicable spectrum regulations and institutional safety requirements.

16.2 Suggested functional hardware

Subsystem	Possible prototype element	Purpose
Audio source	Microphone / audio generator	4 kHz-band-limited message
Audio filter	Active or passive LPF	Reject out-of-band audio
Carrier source	Stable 1 MHz oscillator / generator	RF carrier
AM modulator	Analog multiplier/modulator	Create AM waveform
RF amplifier	Linear RF amplifier	Increase signal level
RF filter	Band-pass / low-pass network	Suppress unwanted products
Mixer	RF mixer	Frequency conversion
LO	1455 kHz oscillator / synthesizer	Superheterodyne conversion
IF filter	455 kHz filter	Channel selectivity
Detector	Diode envelope detector	Recover AM audio
Audio amplifier	Low-power AF stage	Drive speaker/load

16.3 Practical precautions

- Do not assume an RF amplifier is linear merely because the output waveform looks larger; verify spectral cleanliness.
- Use proper grounding, shielding and short RF interconnects to reduce unintended feedback and radiation.
- Keep the audio modulation level controlled so m remains below 1.
- Use a stable reference for the 1 MHz carrier and local oscillator.
- Use a spectrum analyzer, oscilloscope or simulation environment to verify carrier and sideband placement.
- Treat the 10 W carrier value as a design reference; a laboratory prototype may be operated at much lower safe power.

17. ENGINEERING TRADE-OFFS AND LIMITATIONS

17.1 Bandwidth versus complexity

SSB-SC provides the smallest ideal bandwidth among the main alternatives, but it requires more sophisticated generation and detection. AM uses twice the message bandwidth of SSB-SC, yet its envelope detector is simple and robust. For the present educational prototype, the additional 4 kHz of nominal bandwidth is acceptable because the available channel spacing is 20 kHz.

17.2 Power efficiency versus receiver simplicity

Conventional AM is power-inefficient because the carrier consumes most of the transmitted power. Suppressed-carrier schemes improve power utilization, but require carrier recovery. The design therefore accepts the AM power penalty to obtain a simpler receiver and easier demonstration.

17.3 IF/image rejection trade-off

The 455 kHz IF provides a useful fixed-frequency processing stage. However, the mixer creates an image response at 1910 kHz for the selected high-side LO. The receiver therefore requires RF preselection. This is a normal superheterodyne trade-off rather than a design error.

17.4 VSB limitation for this application

VSB is valuable for analog television because it reduces the bandwidth burden of a wide video signal while retaining a controlled vestige. For a narrow 4 kHz voice link, the additional VSB filtering complexity is not justified by the assignment's prototype objective.

17.5 Limitations of the preliminary design

- The calculations use the specified idealized message bandwidth and carrier power.
- The 53-channel result is an idealized count and is not a regulatory allocation plan.
- Actual RF filter transition bands and transmitter spectral masks are not fully specified.
- Receiver sensitivity in dBm is not calculated because noise figure, temperature, required SNR and detector threshold are not provided.
- Image-rejection performance in dB is not calculated because RF filter Q and response are not specified.
- Simulation figures are analytical/design-support results; laboratory measurements should be used for final hardware verification.

18. FINAL ENGINEERING DECISION

WOULD THIS SYSTEM BE DEPLOYED?
YES — for the intended low-power educational voice-broadcast prototype, subject to laboratory power limits, correct filtering and applicable RF regulations.

18.1 Justification

Decision factor	Assessment	Effect on decision
Bandwidth compliance	8 kHz AM occupied bandwidth within 20 kHz channel grid	Positive
Power calculation	12.45 W total AM power for 10 W carrier reference	Acceptable; carrier-heavy
Spectrum utilization	Less efficient than SSB, but adequate for the specified spacing	Acceptable
Detection complexity	Simple envelope detection	Strong positive
Receiver architecture	Superheterodyne gives fixed IF selectivity	Strong positive
IF/image	455 kHz IF; image at 1910 kHz requires RF preselection	Manageable
Noise	AWGN requires filtering and adequate received SNR	Manageable
Implementation cost	Conventional AM chain can be prototyped at low cost	Positive
Voice suitability	AM is well suited to simple voice broadcast demonstrations	Positive
VSB-TV suitability	Not selected for this voice application	Neutral

18.2 What would change for a real deployment?

A production or licensed system would require a formal regulatory frequency allocation, transmitter spectral-mask compliance, antenna-system design, measured receiver sensitivity and selectivity, harmonic-emission testing, electromagnetic compatibility evaluation, thermal design and documented safety procedures. If spectrum efficiency became the dominant requirement, SSB-SC would be a strong redesign candidate; if the application changed to analog television, VSB would become much more relevant.

19. FINAL SPECIFICATION SHEET

Item	Final design specification
Application	Low-power educational analog voice communication link
Message	Voice/audio, 0–4 kHz
Selected modulation	Conventional AM
Modulation index	0.7
Carrier frequency	1 MHz
Carrier power reference	10 W
AM occupied bandwidth	8 kHz
LSB edge	996 kHz
USB edge	1004 kHz
Total sideband power	2.45 W
Total AM power	12.45 W
Nominal RF channel spacing	20 kHz
Idealized channel count	53 intervals across 540–1600 kHz
Transmitter	Audio LPF → AM modulator → RF PA → output filter → antenna
Receiver	RF preselection → mixer → 455 kHz IF → envelope detector → audio amplifier
IF	455 kHz
High-side LO	1455 kHz
Image frequency	1910 kHz
Detection	Envelope detection
Noise model	AWGN / practical receiver limitations
Deployment decision	YES for intended educational prototype

20. CONCLUSION AND REFERENCES

20.1 Conclusion

The communication link has been developed as a complete engineering design rather than as a collection of isolated equations. The 4 kHz voice requirement leads to an 8 kHz conventional-AM occupied bandwidth. With a 1 MHz carrier, the specified signal extends from 996 kHz to 1004 kHz. For a 10 W carrier and modulation index of 0.7, the total sideband power is 2.45 W and the total transmitted AM power is 12.45 W.

Among the candidate modulation methods, conventional AM is selected because its receiver can use simple envelope detection and its bandwidth fits comfortably within the 20 kHz channel grid. A superheterodyne receiver is preferred over a TRF receiver because fixed-IF processing provides more predictable selectivity and gain. With a 455 kHz IF and high-side injection, the local oscillator is 1455 kHz and the image frequency is 1910 kHz, demonstrating the need for RF preselection.

The final engineering decision is YES for the intended low-power educational prototype. The design remains subject to practical filtering, stability, noise and regulatory constraints for any real-world deployment.

20.2 References / source basis

- ECA08 – Analog and Digital Communication, CO1 Design Task Assignment, supplied course document.
- Course-level communication systems theory covering AM, DSB-SC, SSB-SC, VSB-SC, TRF receivers and superheterodyne receivers.
- Analytical calculations and simulation-style figures generated specifically for the design parameters stated in the assignment.

20.3 Final submission checklist

Required element	Included
Communication-system block diagram	YES
RF frequency-allocation plan	YES
Modulation equations and spectra	YES
AM/DSB-SC/SSB-SC/VSB-SC comparison	YES
AM power and bandwidth calculations	YES
Transmitter block diagram	YES
TRF versus superheterodyne comparison	YES
Receiver parameter analysis	YES
IF and image-frequency calculation	YES
Simulation / analytical plots	YES
Verification table	YES
Final engineering recommendation	YES